

PROJECT: Serverless Image/Video Processing

Services: S3, Lambda, Step Functions, Rekognition, SNS

Goal: Automatically process images or videos uploaded to S3 (e.g., resize, label detection).

Skills: Event-driven architecture, serverless orchestration.

Definition: This project implements a serverless image processing pipeline on AWS that automatically analyzes images uploaded to an S3 bucket. The system uses Lambda and Amazon Rekognition to detect objects in the image and send notifications using SNS. AWS Step Functions orchestrates the workflow to ensure smooth execution. The solution is fully automated and event-driven.

Scope of the Project: The scope of this project is to build an automated and scalable image analysis system using AWS serverless services. It focuses on detecting objects in uploaded images and notifying users with the results. The system demonstrates event-driven architecture, AI integration, and cloud automation. It can be extended to include advanced features like storing results in databases or adding decision logic for alerts.

Methodologies:

- 1. Method Used in the Project:** This project uses the Event-Driven Serverless Architecture method.
 - In this approach, an event (image upload to an S3 bucket) automatically triggers AWS services like Lambda, Step Functions, Rekognition, and SNS to process the image and send notifications.
 - Each service runs only when required and communicates through events rather than manual execution.
 - Event-driven serverless architecture is best suited because image processing needs to run only when a new image is uploaded.
 - It provides automatic scaling, reduced operational overhead, and cost efficiency since resources run only when triggered.
 - It also enables seamless integration between AWS services, making the workflow automated, reliable, and highly scalable for real-time image analysis.
- 2. Alternative Method:** Another method that could be used for this project is a traditional server-based architecture using services like EC2 instances to continuously run an application that monitors the S3 bucket for new images and processes them.
 - Another possible approach is a scheduled batch processing method where a scheduled job periodically checks the bucket and processes images in batches.
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Services Used in this Project:

- 1. Amazon S3 (Simple Storage Service)**
 - **Definition:** Amazon S3 is a cloud object storage service used to store and retrieve files such as images, videos, and documents.
 - **Purpose:** In this project, S3 is used to store the uploaded images that will be processed and analyzed.
 - **Role in this Project:** S3 acts as the input source that triggers the workflow when a new image is uploaded.

2. Lambda

- **Definition:** AWS Lambda is a serverless compute service that runs code automatically in response to events without managing servers.
- **Purpose:** In this project, Lambda is used to process the uploaded image and call Rekognition to analyze it.
- **Role in this Project:** Lambda acts as the processing component that reads the image from S3, performs image analysis using Rekognition, and sends the results to SNS and the output bucket.

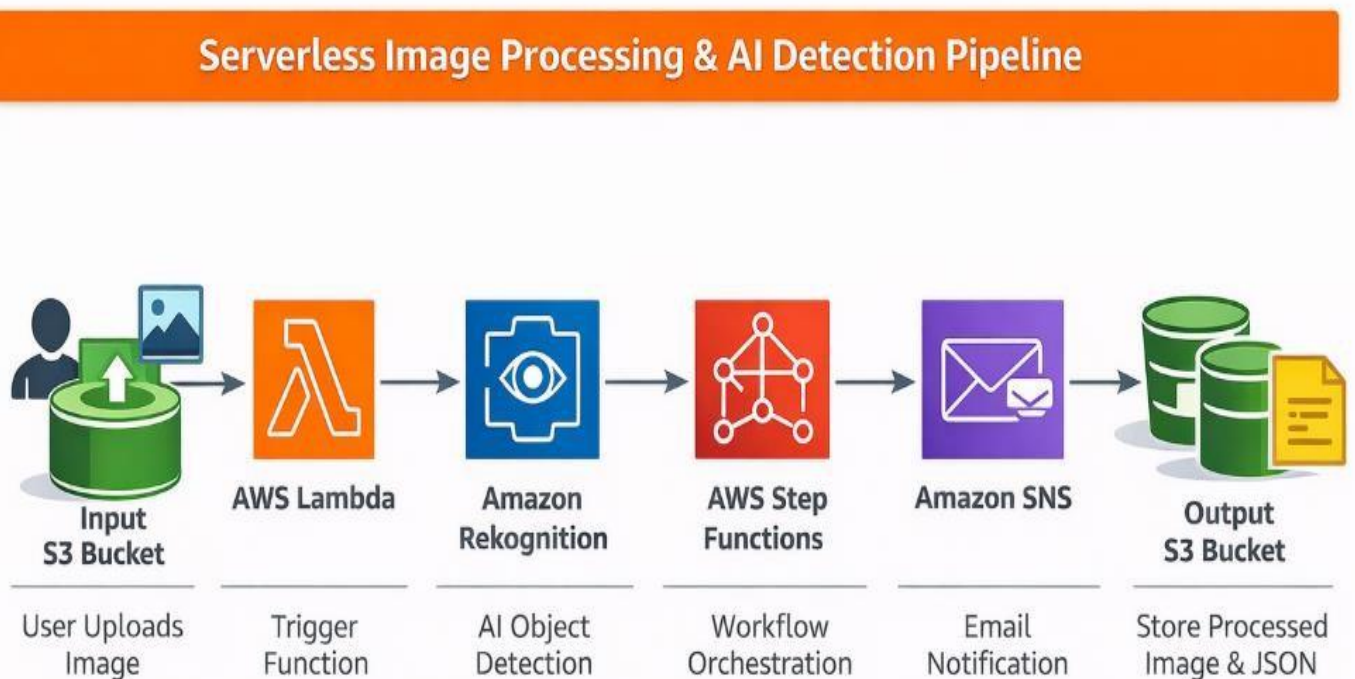
3. Step Function

- **Definition:** AWS Step Functions is a serverless workflow service used to coordinate and manage multiple AWS services in a defined sequence.
- **Purpose:** In this project, Step Functions is used to orchestrate the workflow that triggers Lambda and manages the image processing steps.
- **Role in this Project:** Step Functions acts as the workflow controller that manages and executes the sequence of tasks in the image processing pipeline.

4. Simple Notification Service (SNS)

- **Definition:** Amazon SNS is a cloud messaging service used to send notifications to users through email, SMS, or other endpoints.
- **Purpose:** In this project, SNS is used to send notifications about the image analysis results.
- **Role in this Project:** SNS acts as the notification service that delivers the detected image information to the user via email.

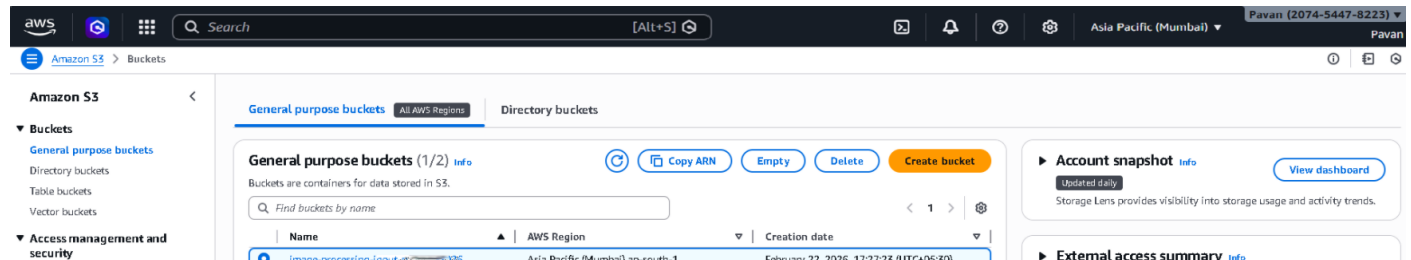
Architecture Diagram:



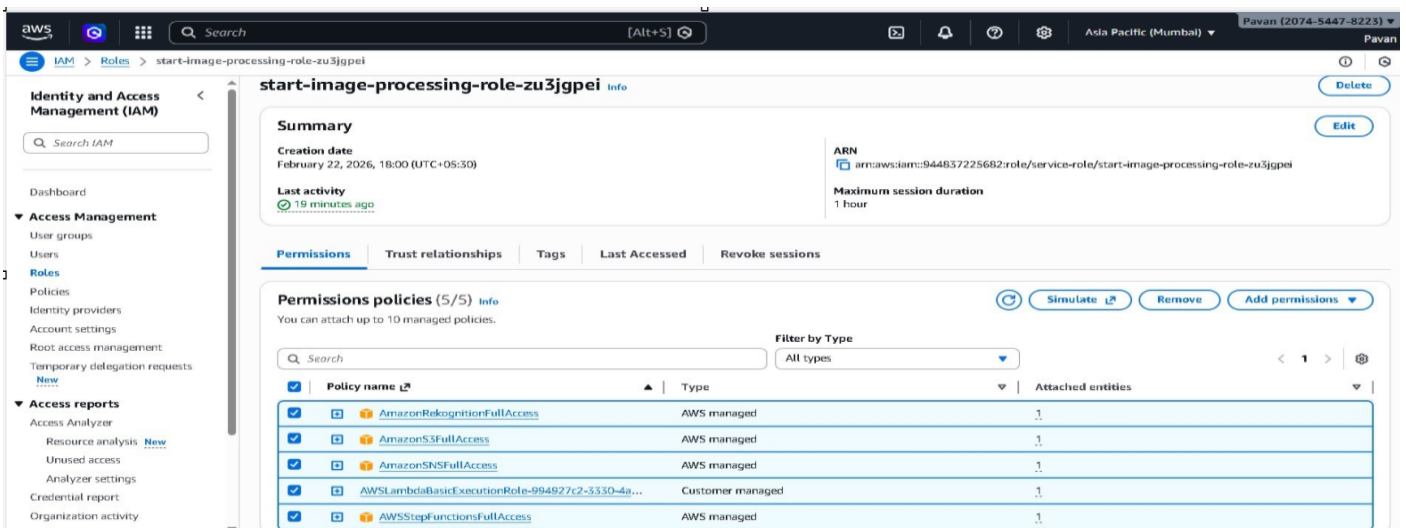
Implementation and Execution of Serverless Image/Video Processing

STEP 1: Created two S3 buckets

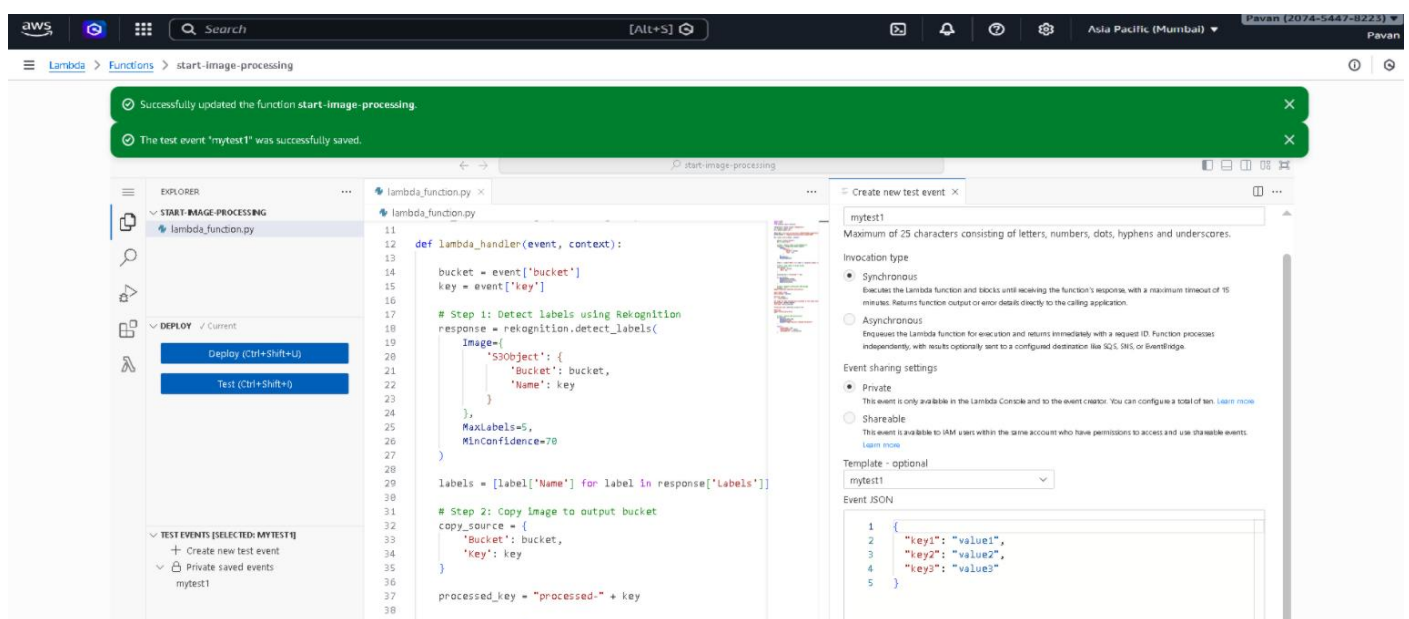
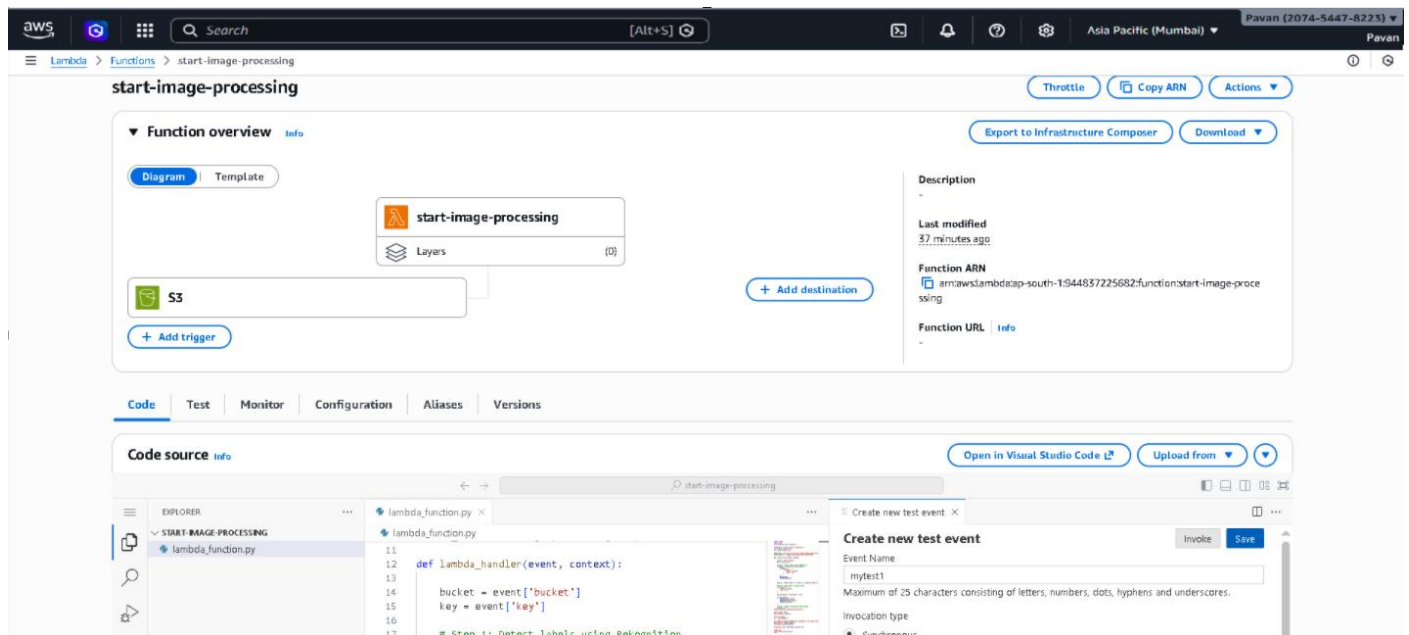
- Input Bucket: To upload images
- Output Bucket : To store processed images and results



STEP 2: Created an IAM role and attached required policies.

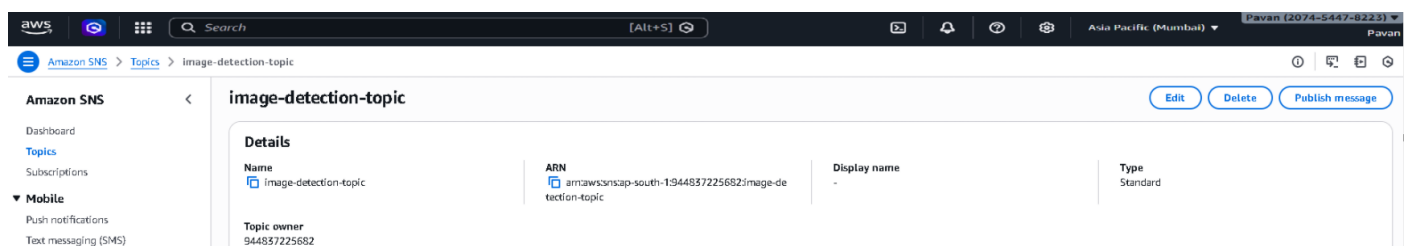


STEP 3: Created a Lambda function and added Python code to read images from the S3 input bucket and process them.



STEP 4: Created an SNS topic and subscription to send email notifications with the image analysis results.

➤ Created Topic



➤ Created Subscription

The screenshot shows the AWS SNS console. The left sidebar has a menu with 'Amazon SNS', 'Topics', 'Subscriptions', and 'Mobile'. The main content area is titled 'Subscription: 252873e1-aafe-43e4-9b4d-580f0d7a3f1e'. It contains a 'Details' section with the following information:

- ARN:** arnaws:sns:ap-south-1:944837225682:image-detection-topic:252873e1-aafe-43e4-9b4d-580f0d7a3f1e
- Endpoint:** sushmabesan9@gmail.com
- Topic:** [image-detection-topic](#)
- Subscription Principal:** arnaws:iam::944837225682:root
- Status:** Confirmed
- Protocol:** EMAIL

There are 'Edit' and 'Delete' buttons in the top right corner.

STEP 5: Created a Step Functions state machine to orchestrate the workflow and invoke the Lambda function.

The screenshot shows the AWS Step Functions console. The left sidebar has a menu with 'Step Functions', 'State machines', 'Execution Inspector', 'Activities', and 'Developer resources'. The main content area is titled 'Execution: image-process-test5'. It shows a green banner at the top stating 'Execution started successfully'. Below this, there are tabs for 'Details', 'Execution input and output', and 'Definition'. The 'Details' tab is active, showing the following information:

- Execution status:** Succeeded
- Execution type:** Standard
- Execution ARN:** arnaws:states:ap-south-1:944837225682:execution:image-processing-workflow-v1:image-process-test5
- IAM role ARN:** arnaws:iam::944837225682:role/service-role/StepFunctions-image-processing-workflow-v1-role-462516a71
- State transitions:** 3
- Start time:** Feb 22, 2026, 22:47:05.769 (UTC+05:30)
- End time:** Feb 22, 2026, 22:47:07.574 (UTC+05:30)
- Duration:** 00:00:01.805
- Alias:** -
- Version:** -

There are buttons for 'Edit state machine', 'New execution', and 'Actions' in the top right corner.

The screenshot shows the AWS Step Functions console with the 'Graph view' selected. The workflow is a simple linear process starting with a 'Start' node, followed by an 'AWS Lambda Invoke' node labeled 'ProcessImage', and ending with an 'End' node. The 'ProcessImage' node is highlighted with a green box. To the right of the graph, there is a 'ProcessImage' section with tabs for 'Input/Output', 'Variables', 'Details', 'Definition', and 'Events'. The 'Input/Output' tab is active, showing the state input and output JSON.

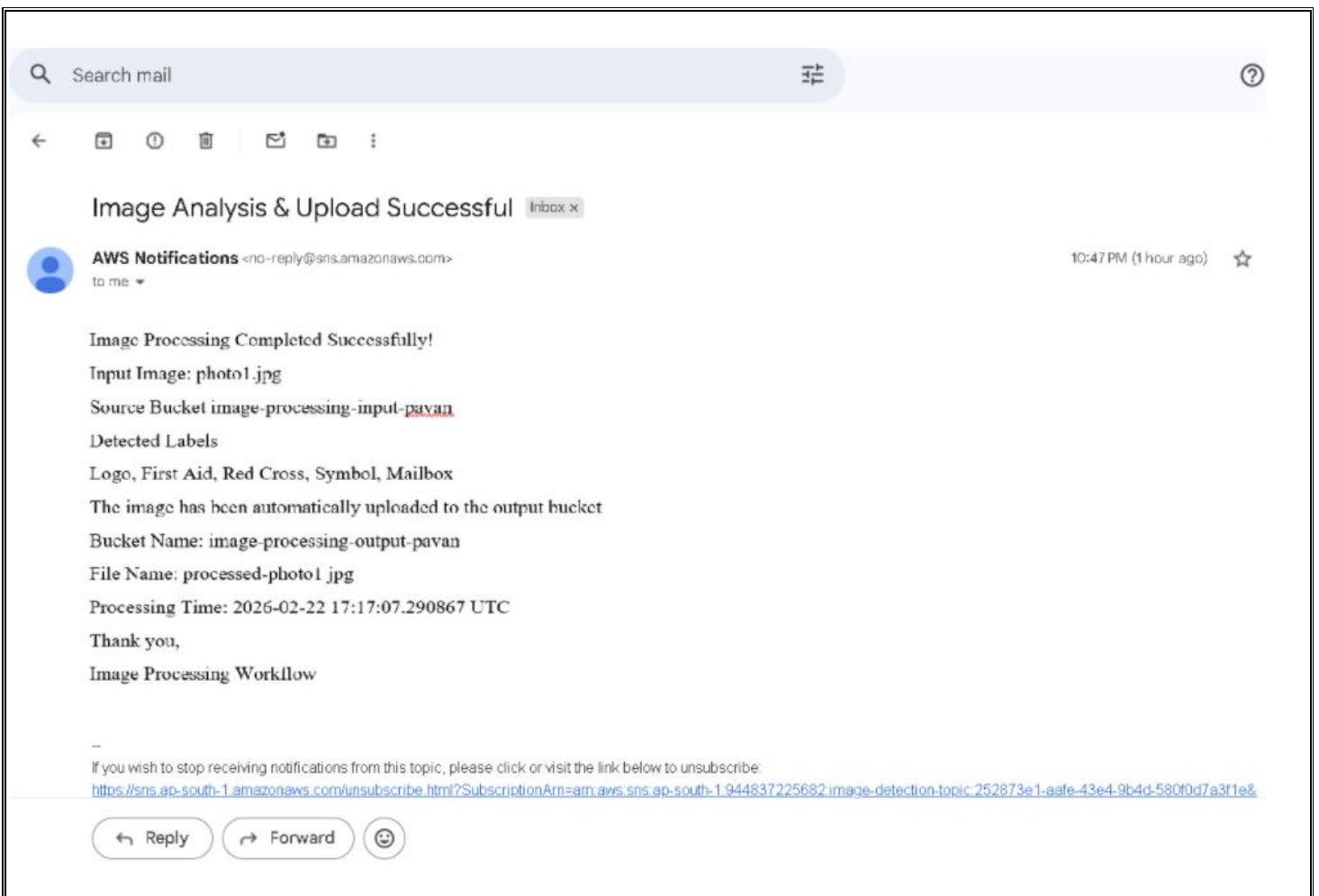
State input:

```
1 {
2   "bucket": "image-processing-input-sushma-2026",
3   "key": "photo1.jpg"
4 }
```

State output:

```
1 {
2   "bucket": "image-processing-input-sushma-2026",
3   "key": "photo1.jpg",
4   "lambdaResult": {
5     "statusCode": 200,
6     "DetectedLabels": [
7       "Logo",
8       "First Aid",
9       "Red Cross",
10      "Symbol",
11      "Mailbox"
12    ],
13   "UploadedFile": "processed-photo1.jpg"
14 }
15 }
16 }
```

STEP 6: Tested the workflow by uploading an image to the S3 input bucket and verifying notification.



Troubleshooting:

1. Execution failed when using the same Step Function execution name
 - **Issue:** Step Functions does not allow the same execution name again.
Solution: Use a unique execution name each time or allow AWS to generate one automatically.
2. Output bucket not showing processed image
 - **Issue:** No image appeared in the output bucket after execution.
Solution: Add **S3 copy or upload code in Lambda** (for example `s3.copy_object` or `s3.put_object`) to store the processed image in the output bucket

Conclusion:

This project demonstrates how AWS serverless services can be used to build an automated image processing system. By integrating S3, Lambda, Rekognition, Step Functions, and SNS, the system analyzes images and sends notifications without managing servers. The workflow is event-driven, scalable, and cost-efficient because it runs only when an image is uploaded. This project highlights the power of cloud automation and AI services in building modern, efficient applications.